

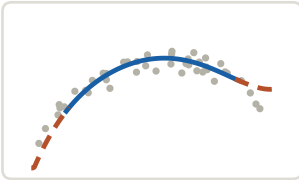
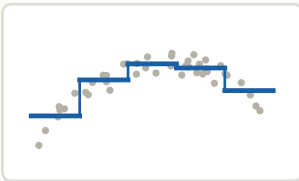
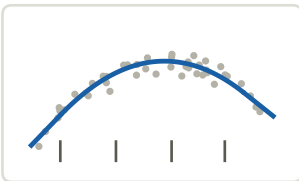
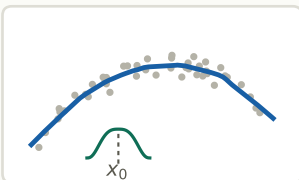
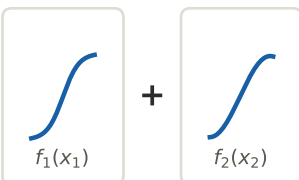
# Beyond linear regression: five ways to fit a curve

Read the shape of the scatterplot, build features of X to match it, then fit a linear model.

$$f(X) = \sum_m \beta_m h_m(X)$$

The shape of the scatterplot suggests which features to build.

pick known functions of X, then fit a linear model.

| Method                         | The fitted shape                                                          | What you fit                                                                        | Trade-off                                                               |                                                             |
|--------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------|
| features of X, then fit linear | <b>Polynomial regression</b><br>One global curve.<br>tune: degree d       |    | Add powers $h_m = X, X^2, \dots, X^d$ ; one curve over all of X.        | Simple, but swings at the tails — weak extrapolation.       |
|                                | <b>Step function</b><br>Bin means.<br>tune: cutpoints                     |    | Bins: $h_m = I(c_k \leq X < c_{k+1})$ ; one constant per bin.           | Easy to read, but misses trends without natural breaks      |
|                                | <b>Natural cubic spline</b><br>Smooth joined pieces.<br>tune: knots / df  |   | Joined cubics; basis $(X - \xi_k)_+^3$ ; straight past the outer knots. | Smooth and stable at edges; you place the knots $\xi_k$ .   |
| local fit                      | <b>Local regression (LOESS)</b><br>Fit from nearby points.<br>tune: span  |  | At each $x_0$ : weighted least squares, weights $K(x_i, x_0)$ .         | Very flexible, but weak for many predictors ( $p \leq 4$ ). |
| sum of smooths                 | <b>Generalized additive (GAM)</b><br>Sum of smooths.<br>tune: df per term |  | $y = \beta_0 + \sum_j f_j(X_j)$ ; fit each $f_j$ by backfitting.        | Flexible and readable, but additive — add interactions.     |

## Why they belong together

All five keep interpretability: transform or segment X, then fit linear. LOESS fits locally; a GAM sums smooths. More flexibility can cost stable inference — the tuning knob sets the balance. *The scatterplot's shape is your guide to which method to reach for.*